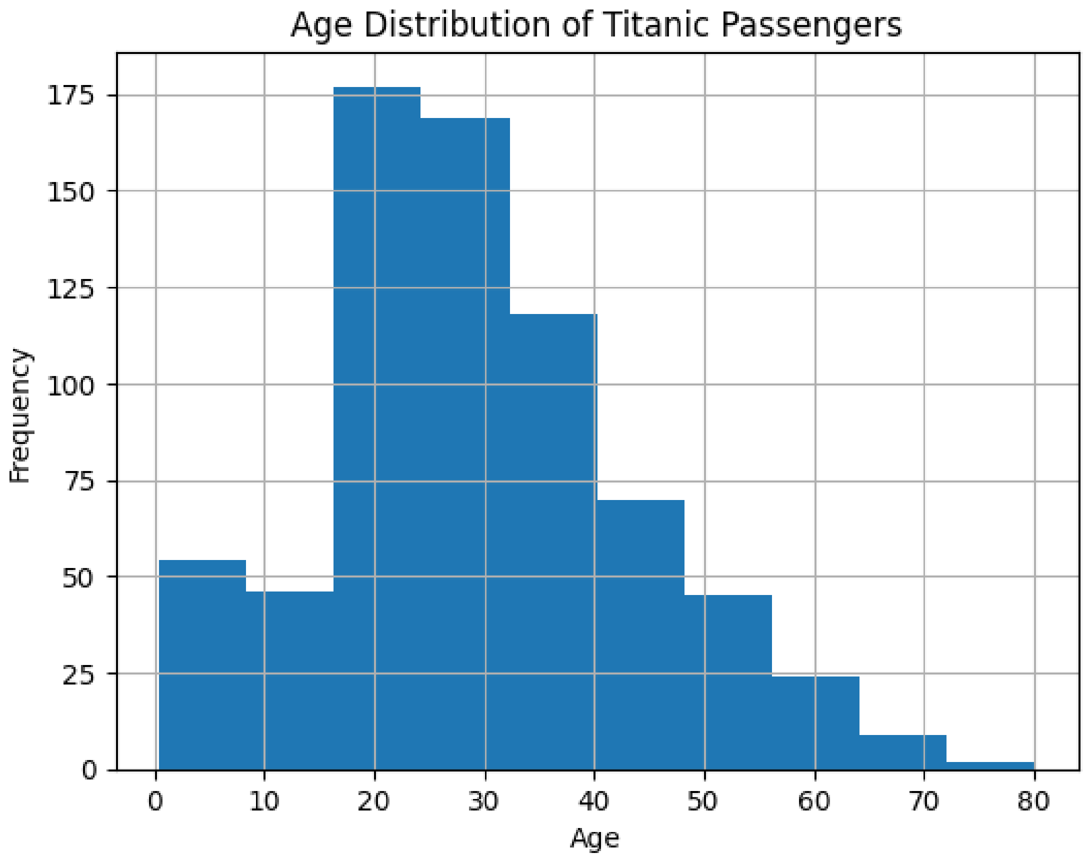
```python
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("/content/titanic.csv")

print(df.info())
print(df.describe())
```

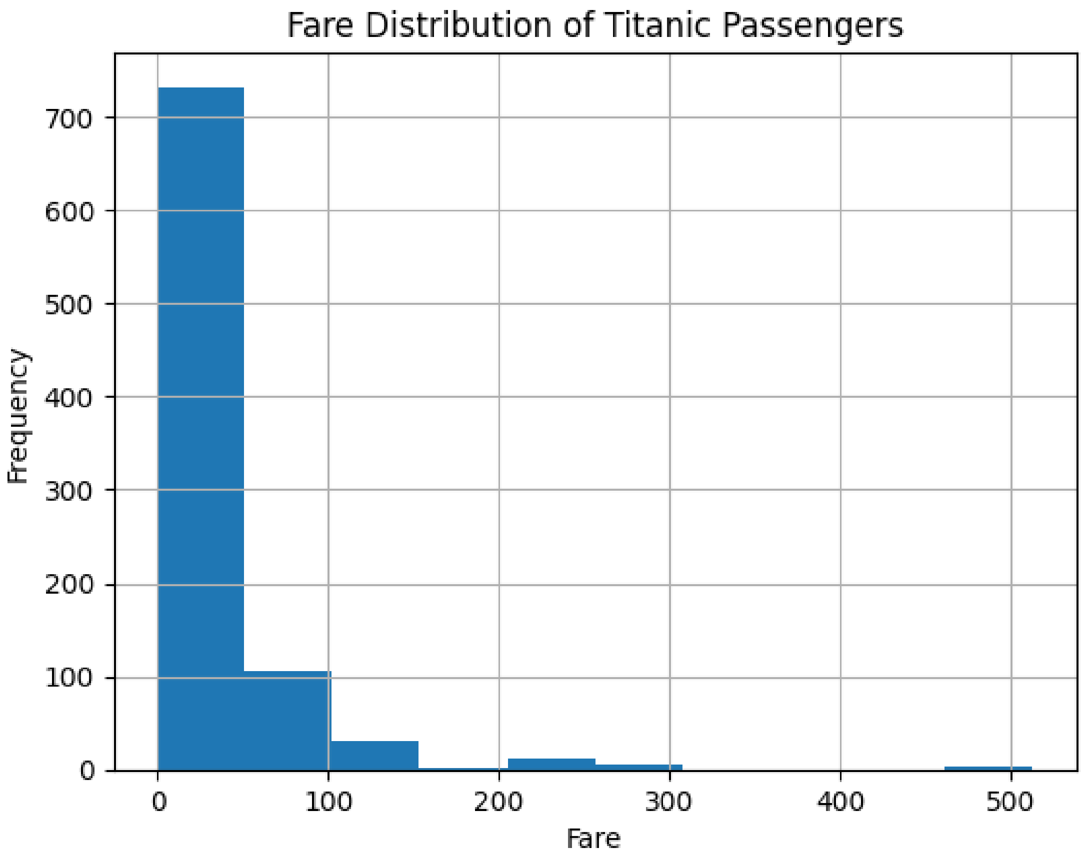
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype  
---  ------       --------------  -----  
 0   PassengerId  891 non-null    int64  
 1   Survived     891 non-null    int64  
 2   Pclass       891 non-null    int64  
 3   Name         891 non-null    object 
 4   Sex          891 non-null    object 
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64  
 7   Parch        891 non-null    int64  
 8   Ticket       891 non-null    object 
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    object 
 11  Embarked     889 non-null    object 
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
None
       PassengerId    Survived      Pclass         Age       SibSp  \
count   891.000000  891.000000  891.000000  714.000000  891.000000   
mean    446.000000    0.383838    2.308642   29.699118    0.523008   
std     257.353842    0.486592    0.836071   14.526497    1.102743   
min       1.000000    0.000000    1.000000    0.420000    0.000000   
25%     223.500000    0.000000    2.000000   20.125000    0.000000   
50%     446.000000    0.000000    3.000000   28.000000    0.000000   
75%     668.500000    1.000000    3.000000   38.000000    1.000000   
max     891.000000    1.000000    3.000000   80.000000    8.000000   

            Parch        Fare  
count  891.000000  891.000000  
mean     0.381594   32.204208  
std      0.806057   49.693429  
min      0.000000    0.000000  
25%      0.000000    7.910400  
50%      0.000000   14.454200  
75%      0.000000   31.000000  
max      6.000000  512.329200  
```

```python
plt.figure()
df['Age'].dropna().hist()
```

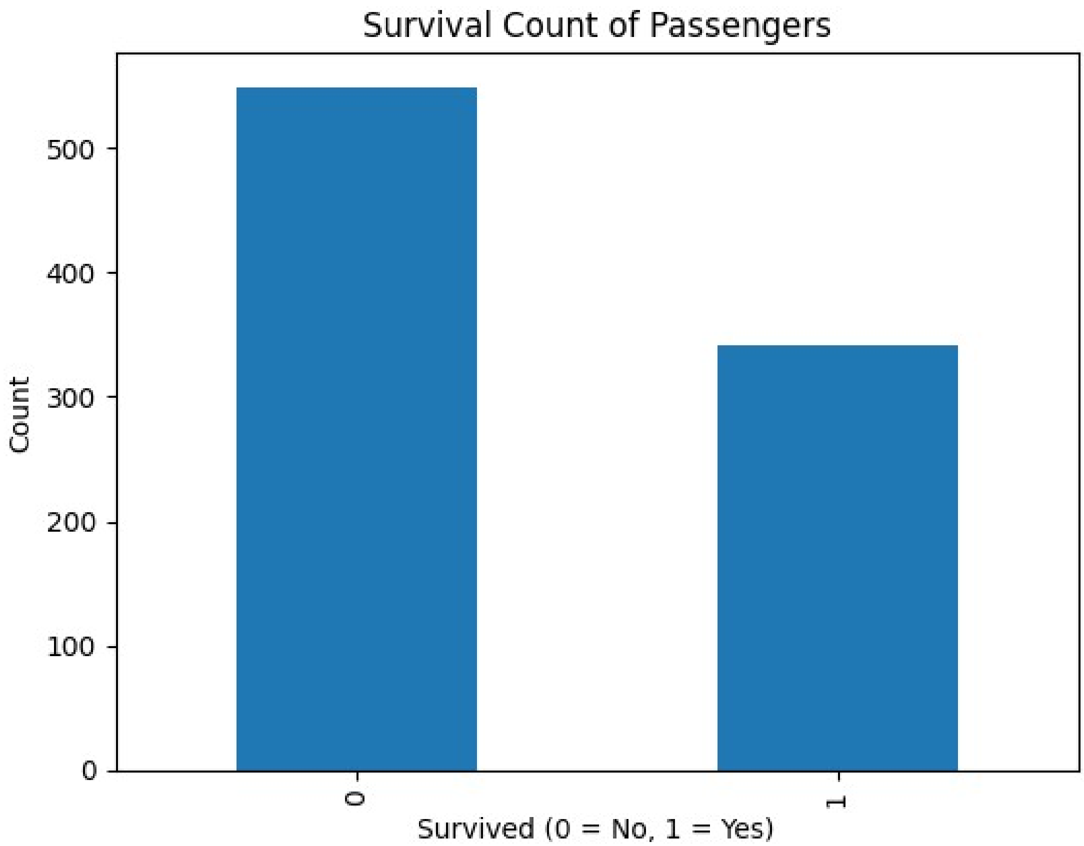
```
plt.xlabel("Age")
plt.ylabel("Frequency")
plt.title("Age Distribution of Titanic Passengers")
plt.show()
```



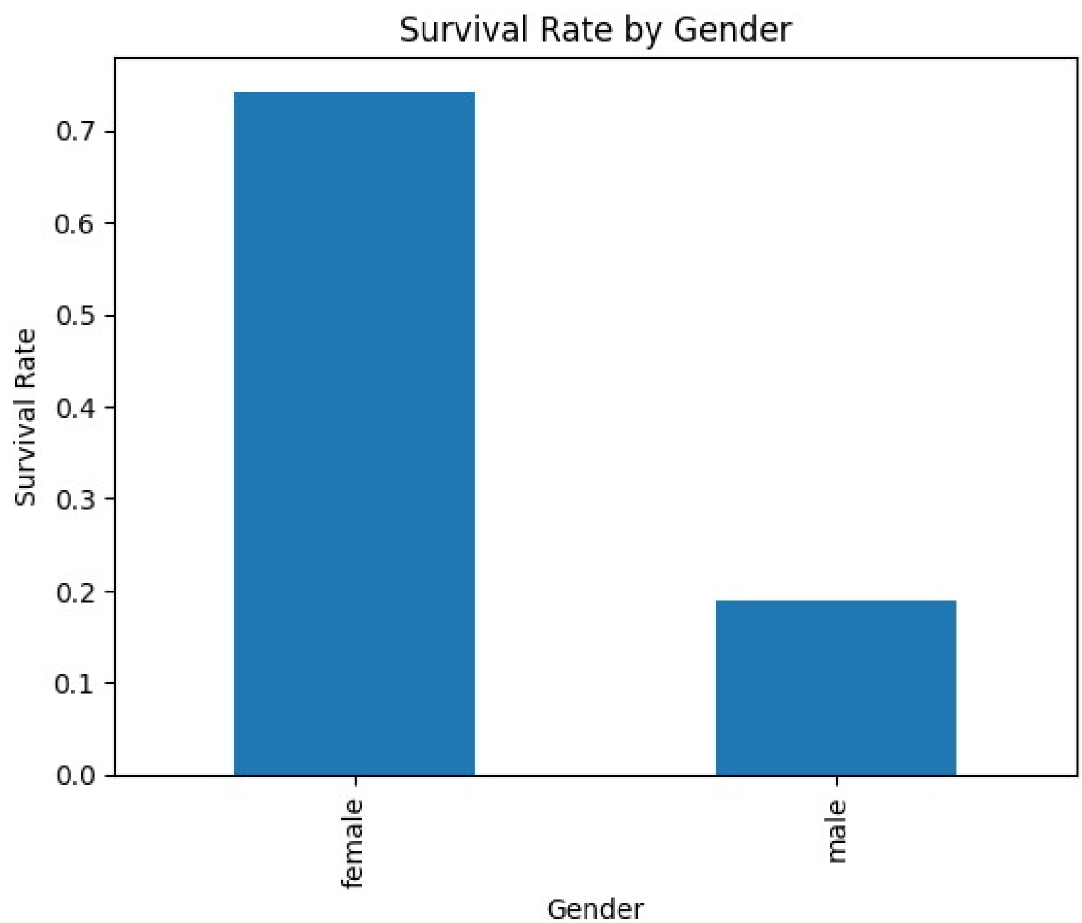

```
plt.figure()
df['Fare'].hist()
plt.xlabel("Fare")
plt.ylabel("Frequency")
plt.title("Fare Distribution of Titanic Passengers")
plt.show()
```



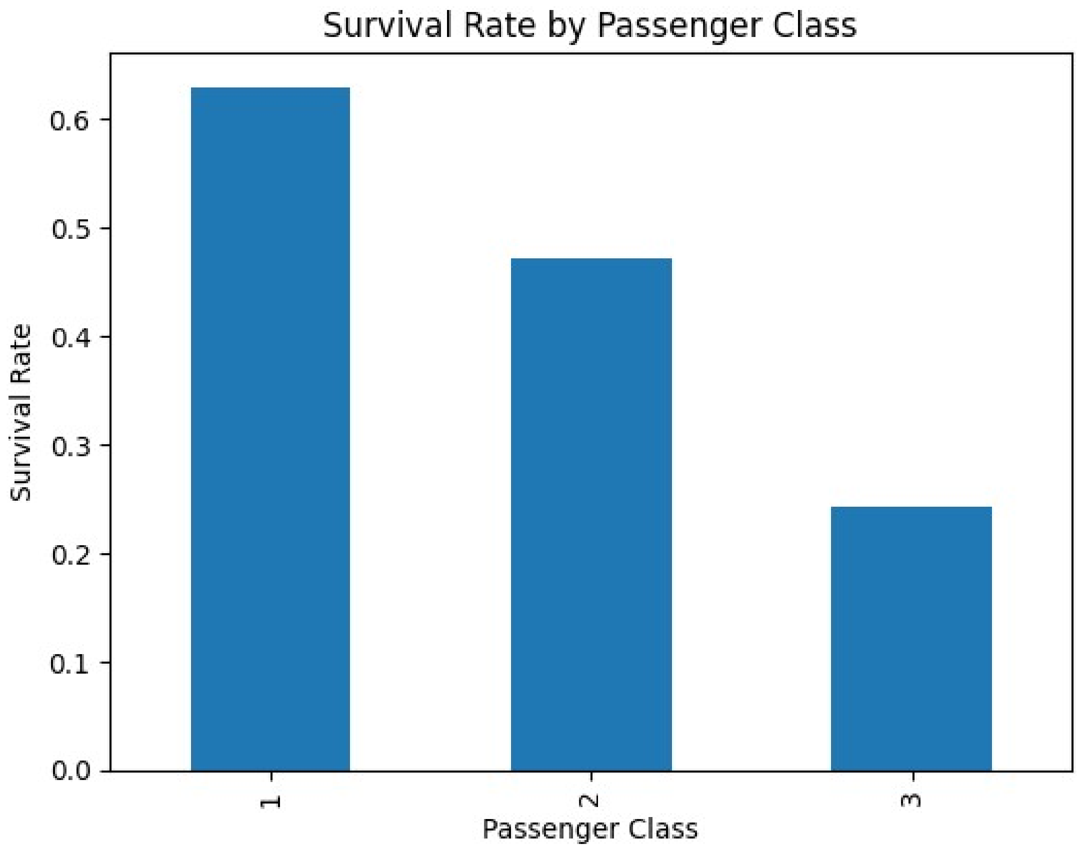

```
plt.figure()
df['Survived'].value_counts().plot(kind='bar')
plt.xlabel("Survived (0 = No, 1 = Yes)")
plt.ylabel("Count")
plt.title("Survival Count of Passengers")
plt.show()
```




```
gender_survival = df.groupby('Sex')['Survived'].mean()

plt.figure()
gender_survival.plot(kind='bar')
plt.xlabel("Gender")
plt.ylabel("Survival Rate")
plt.title("Survival Rate by Gender")
plt.show()
```




```
class_survival = df.groupby('Pclass')['Survived'].mean()

plt.figure()
class_survival.plot(kind='bar')
plt.xlabel("Passenger Class")
plt.ylabel("Survival Rate")
plt.title("Survival Rate by Passenger Class")
plt.show()
```



Survival Rate by Passenger Class
Survival Rate
0.0
0.1
0.2
0.3
0.4
0.5
0.6
1
2
3
Passenger Class